CHAOLIN TIAN

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Research Interests

My research examines how knowledge workers develop their careers and create innovation. Focusing on scientists, I study how recognition and social capital interact to shape career opportunities, mobility, and collaboration. I also explore how AI may change these relationships.

Academic Positions

Northwestern University (NU) Oct 2025 – Present
Postdoctoral Fellow, Kellogg School of Management *Evanston*
Supervisor: Prof. Brian Uzzi

Southern University of Science and Technology (SUSTech) Jul 2025 – Sep 2025
Visiting Research Fellow, Department of Statistics and Data Science *Shenzhen*

Education

Southern University of Science and Technology (SUSTech) Sep 2021 – Jun 2025
Ph.D. in Data Science, Department of Statistics and Data Science *Shenzhen*
Advisor: Associate Prof. Yifang Ma
Dissertation: "Statistical Analysis and Modeling of Scholar Academic Characteristics Based on Large-scale Research Data"

Southern University of Science and Technology (SUSTech) Sep 2019 – Jun 2021
M.S. in Mathematics, Department of Statistics and Data Science *Shenzhen*
Advisor: Prof. Guo-Liang Tian
Thesis: "Comparisons of Multiple Independent Groups for Continuous Proportional Data"

Yunnan Normal University (YNNU) Sep 2015 – Jun 2019
B.A. in Statistics, School of Mathematics, Rank: 1/40 *Kunming*

Working Papers and Manuscripts

Working Papers

- **Chaolin Tian**, & Brian Uzzi. (2026). *From Academia to Industry: How Career Transitions Reshape Scientific Collaboration*.
- **Chaolin Tian**, Zhu Wang, Yifang Ma, & Brian Uzzi. (2026). *Artificial Intelligence and the Changing Role of Social Capital in Science*.
- **Chaolin Tian**, Yifang Ma, & Brian Uzzi. (2025). *The Universal Dynamics of Exceptional Achievement*.
- Moh Hosseinioun, **Chaolin Tian**, & Brian Uzzi. (2026). *The Long-Run Career Consequences of MBA Admission and Enrollment Decisions*.
- Ningyuan Shen, **Chaolin Tian**, & Yifang Ma. (2026). *The Protective Role of Scientific Recognition in Scientific Exploration*.

Manuscripts

- **Chaolin Tian**, Brian Uzzi, & Yifang Ma. (2025). *Unveiling the Intellectual Dimensions of the Innovativeness Metrics in Science*.
- **Chaolin Tian**, & Yifang Ma. (2025). *The Innovation Trap of Early-Career Researchers*.
- **Chaolin Tian**, Fan Jiang, Yurui Huang, Cassidy R. Sugimoto, & Yifang Ma. (2024). *Prize Recognition Boosts Women's Scientific Status and Career Longevity More than Men's*.
- Yurui Huang, Langtian Ma*, **Chaolin Tian**, Xunyi Jiang*, Roberta Sinatra, & Yifang Ma. (2024). *Quantifying the hierarchical scales of scientists' mobility. arXiv preprint*, arXiv:2401.06794.

Peer-Reviewed Publications

[*] *indicates students I have mentored.*

[1] **Chaolin Tian**, Xunyi Jiang[*], Yurui Huang, Langtian Ma[*], & Yifang Ma. (2026). *Mapping university prestige and hierarchy in China via faculty hiring networks of internationally active Ph.D.s.* **Humanities and Social Sciences Communications**, 13, 379.

[2] **Chaolin Tian**, Yurui Huang, Ching Jin, Yifang Ma, & Brian Uzzi. (2025). *The distinctive innovation patterns and network embeddedness of scientific prizewinners.* **Proceedings of the National Academy of Sciences**, 122(40), e2424143122.

[3] Shibing Xiang, Xin Jiang, Bing Liu, Yurui Huang, **Chaolin Tian**, & Yifang Ma. (2025). *SciConNav: Knowledge navigation through contextual learning of extensive scientific research trajectories.* **Journal of the Association for Information Science and Technology**, 76(10), 1308–1339.

[4] Yurui Huang, Xuesen Cheng, **Chaolin Tian**, Xunyi Jiang[*], Langtian Ma[*], & Yifang Ma. (2024). *Talent hat, cross-border mobility, and career development in China.* **Quantitative Science Studies**, 5(3), 516–532.

[5] Yongshen He[*], Yurui Huang, **Chaolin Tian**, Shibing Xiang, & Yifang Ma. (2024). *Neural embeddings of scientific mobility reveal the stratification of institutions in China.* **Information Processing & Management**, 61(3), 103702.

[6] Yurui Huang, Zimo Wang, **Chaolin Tian**, & Yifang Ma. (2024). *Community detection on elite mathematicians' collaboration network.* **Journal of Data and Information Science**, 9(4), 1–23.

[7] Yurui Huang, **Chaolin Tian**, & Yifang Ma. (2023). *Practical operation and theoretical basis of difference-in-difference regression in science of science: The comparative trial on the scientific performance of Nobel laureates versus their coauthors.* **Journal of Data and Information Science**, 8(1), 29–46.

Invited Talks and Conference Presentations

Invited Talks

- Yunnan Normal University, Kunming, China — Aug 2025
  The Innovative Distinctiveness of Prizewinners and Their Networks
- University of Copenhagen, Copenhagen, Denmark — Jun 2025
  The Innovative Distinctiveness of Prizewinners and Their Networks

Conference Presentations

- International Conference on the Science of Science and Innovation (ICSSI), Copenhagen, Denmark — Jun 2025
  Quantifying the Innovativeness of Celebrated Scientists and Their Embeddedness in Collaboration Networks
- The 7th Tianfu Forum on Scientific Metrology and Technology Evaluation, Chengdu, China — Dec 2024
  Quantifying the Innovativeness of Celebrated Scientists and Their Embeddedness in Collaboration Networks
- SUSTech–UIC Joint Graduate Academic Symposium — Nov 2024
  Statistical Analysis and Modeling of Scholars' Academic Characteristics Using Large-Scale Research Data
- The 19th Chinese Conference on Complex Networks (CCCN), Guilin, China — Nov 2023
  Quantifying Hierarchy and Dynamics in Chinese Faculty Hiring Networks

Teaching Experience

Introduction to Data Science (STA217)	Fall 2022
Mathematical Statistics (STA5002)	Fall 2021
Computational Statistics (MAT7105)	Spring 2021
Probability and Statistics (MA212)	Spring 2020
Ordinary Differential Equations (MA201a)	Fall 2019

Mentoring Experience

Publication numbers refer to the entries in the Peer-Reviewed Publications section.

Master's Student: Yongshen He (SUSTech), mentored in matching and embedding-based analysis. *Resulting publication:* [5].

Undergraduate Students: Ningyuan Shen, Xunyi Jiang, and Langtian Ma (SUSTech), mentored in data analysis, visualization, and academic writing. *Resulting publications:* [1], [2], [4].

Fellowships and Awards

Outstanding Graduate Student (SUSTech)	Jan 2024
Doctoral Research Fellowship (SUSTech)	Sep 2021 – Jun 2025
Graduate Research Fellowship (SUSTech)	Sep 2019 – Jun 2021
Outstanding Undergraduate Graduate of Yunnan Province (YNNU)	Jul 2019
National Scholarship (YNNU)	Nov 2018
National Endeavor Scholarship (YNNU)	Nov 2017
National Second Prize, China Undergraduate Mathematical Contest in Modeling (YNNU)	Nov 2017

Professional Service

Ad Hoc Reviewer: *Humanities and Social Sciences Communications*; *Scientific Data*; *Discover Education*

Conference Volunteer: Organizing Team Member for three consecutive years, Shenzhen International Conference on Frontiers of Statistics and Data Science (2021–2024)

References

Brian Uzzi, Professor, Kellogg School of Management, Northwestern University.
Email: uzzi@kellogg.northwestern.edu

Yifang Ma, Associate Professor, Department of Statistics and Data Science, Southern University of Science and Technology.
Email: mayf@sustech.edu.cn

Guoliang Tian, Professor, Department of Statistics and Data Science, Southern University of Science and Technology.
Email: tiangl@sustech.edu.cn